Supply Chain Uncertainty, Energy Prices, and Inflation

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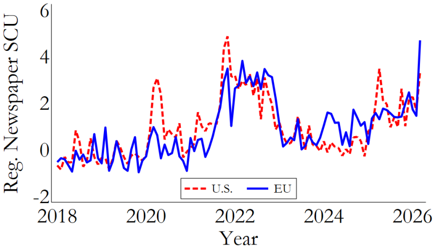
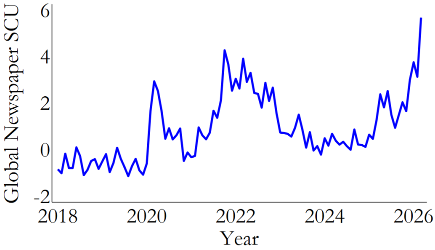
Supply chain uncertainty on the rise



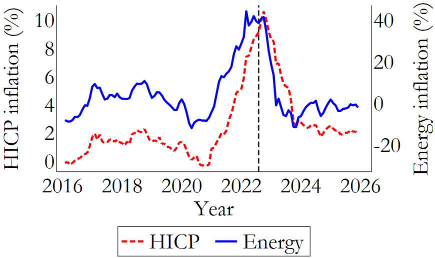
(a) U.S. and EU



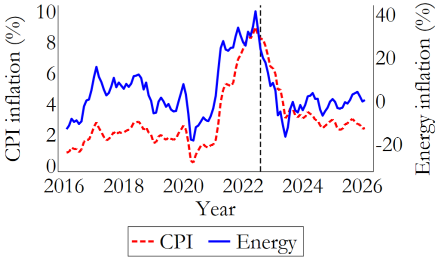
(b) Global

High supply chain uncertainty (SCU) in 2021-2023 and in Feb/Mar 2026

Energy and CPI Inflation in the U.S. and the Euro area



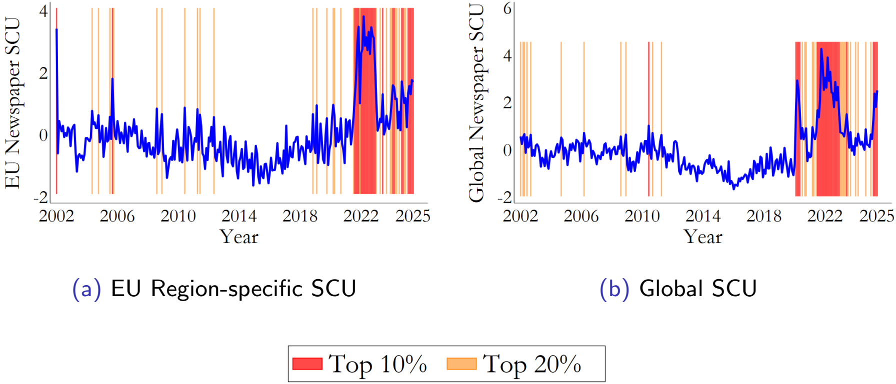
Euro area



U.S.

In 2021 energy and CPI inflation accelerate in tandem

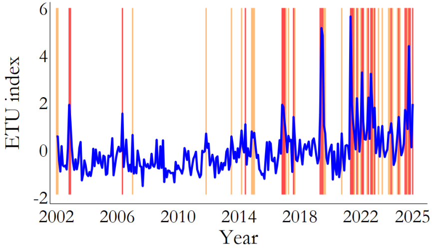
Heightened uncertainty in global supply chains (I)



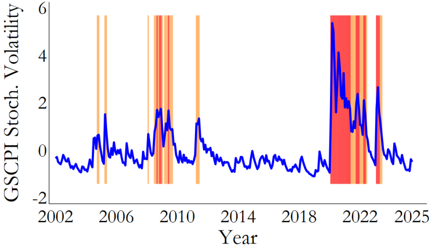
(a) EU Region-specific SCU (b) Global SCU

EU region-pecific SCU: share of articles jointly mentioning supply-chain *and* uncertainty terms in EU economic newspapers → *perceived* SCU.
Global SCU: SCU in global media.

Heightened Uncertainty in Global Supply Chains (II)



(c) ETU index



(d) GSCPI Volatility

ETU: narrow *energy-transport* disruptions in global media.
GSCPI Volatility: estimate of **stochastic-volatility** model of GSCPI $\rightarrow$ *fundamental* unpredictability of supply chains.

SCU measures: construction & keywords

1. **Region-specific SCU** *(baseline; perception, regional)*
   Factiva U.S. / EU newspapers. Share of articles jointly mentioning **supply chain** (*supply chain, bottleneck, disruption, shortage, shock, . . .*; 23 terms, Hassan et al. 2025) **and uncertainty** (*risk, hazard, threat, volatility, unpredictable, . . .*; ˜130 synonyms).

2. **Global SCU** *(perception, worldwide)*
   Same dictionaries, applied to **global** Factiva newspapers. Broader geographic scope.

3. **ETU index** *(Morão 2025; narrow energy-transport)*
   Three dictionaries within **5-word proximity**: **Energy** (*oil, gas, petroleum*) $\times$ **Transportation** (*pipeline, shipping, rail*) $\times$ **Uncertainty** (*threat, risk, concern*). Tracks narrow energy-transport episodes (Gulf of Oman 2019, Colonial Pipeline 2021).

4. **GSCPI Volatility** *(objective, fundamental)*
   No keywords. Stochastic-volatility model on GSCPI (Benigno et al. 2022); filtered conditional variance $\exp(2\sigma_{\psi,t})$ of GSCPI innovations.

Pass-through of energy price shocks to inflation

Local Projections: Oil News Shocks

- Low and high **supply chain uncertainty** regimes.

$$D_t = \mathbf{1}\{SCU_t > p_{80}\}.$$

where $D_t = 1$ when supply chain uncertainty is in the top 20% of its distribution (**Jan. 2002–Jun. 2025**).

- Baseline measure is SCU_t (**regional SCU**).

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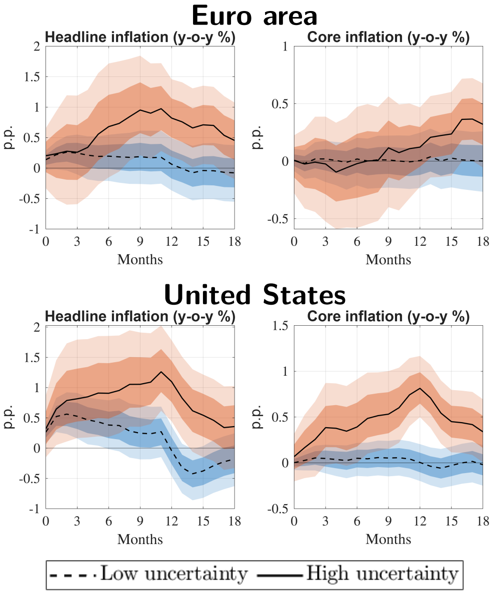
- Baseline measure is SCU_t (**regional SCU**).
- Estimate **state-dependent local projections**:

$$\pi_{t+h} = \alpha_h + \beta_h \varepsilon_t^{\text{oil}} + \beta_h^H \underbrace{\left(D_t \cdot \varepsilon_t^{\text{oil}}\right)}_{\text{interaction}} + \sum_{j=1}^{J} \gamma_{j,h} \pi_{t-j} + \nu_{t+h}$$

- Dummy D_t interacted with high-frequency **oil supply news** surprises $\varepsilon_t^{\text{oil}}$ (Känzig, 2021).

Local Projections: Oil News Shocks and Inflation



Pass-through to inflation of **oil** news shocks significantly **larger** conditional on high **SCU**

Robustness

Robustness of pass-through results

1. Across **SCU measures**
2. Dumming-out **Covid** inflation period
3. No role of other measures of **uncertainty** (VIX, Ec Pol uncertainty)
4. No state dependence with **gas shocks**
5. **Top-20** cutoff for "high uncertainty"
6. High pass-through not due to unusually **large** shocks

Main idea:

supply-chain uncertainty amplifies pass-through of energy price shocks to inflation

Core Mechanism I

- Firms use **two inputs**:

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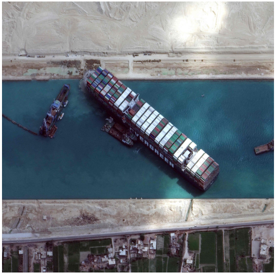

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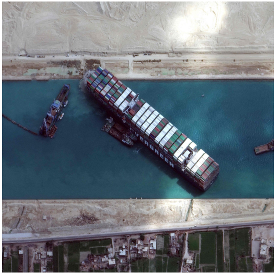

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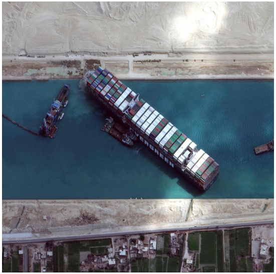

- $\Rightarrow$ **Asymmetry**:
  - **energy** remains available via **liquid** local markets at **premium** price p_E
  - **EV batteries** face stochastic physical delivery **delay**

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- Firms **perceive** larger marginal costs:

  ⇒ **Static** pass-through ⇒ **price overreaction**

  ⇒ **Dynamic** pass-through → **Propagation**

  - Mechanism **scales** with **SCU** (variance of transport. shock)

Strait of Hormuz, Feb–Apr 2026

- **Shock:** U.S.–Israel strikes on Iran (Feb 28) $\Rightarrow$ Hormuz **closed**; tanker traffic $-\mathbf{90}\%$.

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- **Energy (commoditized):** oil $+$10–50%; spot markets **clear at premium**.
- **Specialized inputs: delivery delays, not prices**
  - **Fertilizer** ($\sim$1/3 of seaborne trade via Hormuz): 3–4 Mt/month stranded; urea $+$19–28%; UN emergency exemption Mar 27 for planting season.
  - **Gas inputs for semiconductor industry** (Qatar $\sim$98% of Gulf exports): helium supply $-$35–40%; Cape rerouting $+$19 days for precision equipment. $\Rightarrow$ 19 extra shipping days, 90% flow collapse

Look through supply shocks?

- **Look-through** doctrine based on supply shocks being **transitory**
- Under high SCU $\rightarrow$ Temporary oil shocks **amplified** and endogenously **persistent**
- Monetary policy response should be conditional on degree of SCU

Information value of energy prices: Kalman filter estimation

Information role of energy prices: Kalman filter estimation

State (latent transportation delay):

$$\psi_t = \rho_\psi \psi_{t-1} + \varepsilon_{\psi,t}, \qquad \varepsilon_{\psi,t} \sim \mathcal{N}(0, \sigma_\psi^2)$$

Signals (observation equation):

$$\underbrace{\begin{bmatrix} \text{BDI}_t \\ \pi_t^E \end{bmatrix}}_{\text{noisy signals}} = \underbrace{\begin{bmatrix} \alpha_{\text{BDI}} \\ \alpha_E \end{bmatrix}}_{\text{loadings}} \psi_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}\left(0, \begin{bmatrix} \sigma_{\text{BDI}}^2 & 0 \\ 0 & \sigma_E^2 \end{bmatrix}\right)$$

- $\alpha_{BDI} \equiv$ **loading on BDI (Baltic Dry Index)**.
- $\alpha_E \equiv$ loading on **energy inflation** $\Rightarrow$ How much **energy** is estimated to signal the **transportation shock ("delay")** ψ_t on top of freight BDI index.

State dependency: high vs low supply chain uncertainty

- Divide the sample in **two regimes**: **high** vs **low** supply chain uncertainty based on D_t

$$D_t = \mathbf{1}\{SCU_t > p_{80}\},$$

where $D_t = 1$ when supply chain uncertainty is in the top 20% of its distribution (Jan. 2002–Jun. 2025).

- **Estimate model parameters via MLE**.

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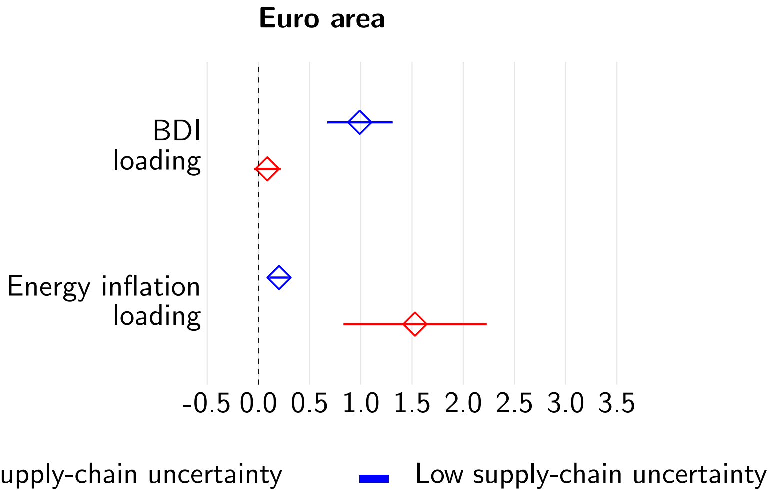
- **Estimate model parameters via MLE**.

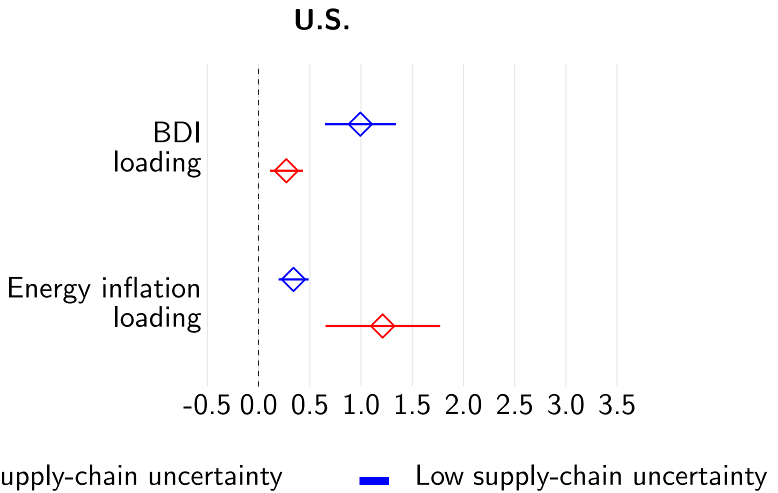
$\Rightarrow$ **Intuition**: measure whether **energy** prices have a particularly high **information content** in **high supply uncertainty** states.

Kalman filter: signal loadings and uncertainty



Result: loading $\alpha_E \geq \alpha_{BDI}$ conditional on **high** supply chain uncertainty

Kalman filter: signal loadings and uncertainty



Parameter estimates

Takeaway from Kalman filter estimates

- **Main idea** When capacity constraint is binding, market prices for transportation no longer serve as reliable indicators of delivery times
- $\Rightarrow$ **Rationing** occurs through **physical capacity** rather than through **price** adjustment.

Evidence from Firms Earnings Calls

Evidence from Earnings Calls

- **Data**: earnings call transcripts of publicly listed U.S. and Euro area firms from NL Analytics (2002Q1–2025Q2). Construct two quarterly indicators:
  - $w_{SC,t} \equiv$ share of sentences with a **supply chain** keyword
  - $w_{E,t} \equiv$ share of sentences with an **energy** keyword
- Test the **conditional co-movement** of the two indicators and its **state dependence** with respect to supply chain uncertainty:

$$\underbrace{w_{SC,t}}_{\text{supply chain words}} = \alpha + \beta_1 \pi_{E,t} + \beta_2 \underbrace{w_{E,t}}_{\text{energy words}} + \beta_3 \left[SCU_t \times w_{E,t} \right] + \varepsilon_t$$

Idea: if firms use energy prices as a *signal* about supply bottlenecks $\Rightarrow$ Expect $\beta_2 > 0$ (co-movement *beyond* energy inflation) and $\beta_3 > 0$ (amplification when SCU is high).

Evidence from Earnings Calls: Results

(a) U.S.

	Supply chain keyword		
	(1)	(2)	(3)
Energy keyword	0.537***	0.680***	0.373***
	(0.088)	(0.072)	(0.067)
Energy inflation		0.043***	0.023***
		(0.006)	(0.005)
SCU × Energy keyword			0.447***
			(0.056)
Adj. R^2	0.280	0.555	0.737
N	94	94	94

(b) Euro area

	Supply chain keyword		
	(1)	(2)	(3)
Energy keyword	0.647***	0.608***	0.478***
	(0.079)	(0.064)	(0.058)
Energy inflation		0.047***	0.011
		(0.007)	(0.008)
SCU × Energy keyword			0.370***
			(0.059)
Adj. R^2	0.413	0.621	0.733
N	94	94	94

Notes: Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Quarterly data, 2002Q1–2025Q2. Earnings calls variables are standardized.

⇒ $\beta_2 > 0$: energy words co-move with supply chain words **beyond** energy inflation

⇒ $\beta_3 > 0$: co-movement is **stronger under high SCU**

Summary of empirical evidence for the EA and the U.S.

- In high **supply-chain uncertainty** states:

1. **Pass-through** of energy price shocks to inflation is **higher**
2. **Information** value of **energy prices** as **signals** increases
3. Firms jointly **mention** "supply chain" and "energy" words systematically

A Stylized Model of the Supply Chain

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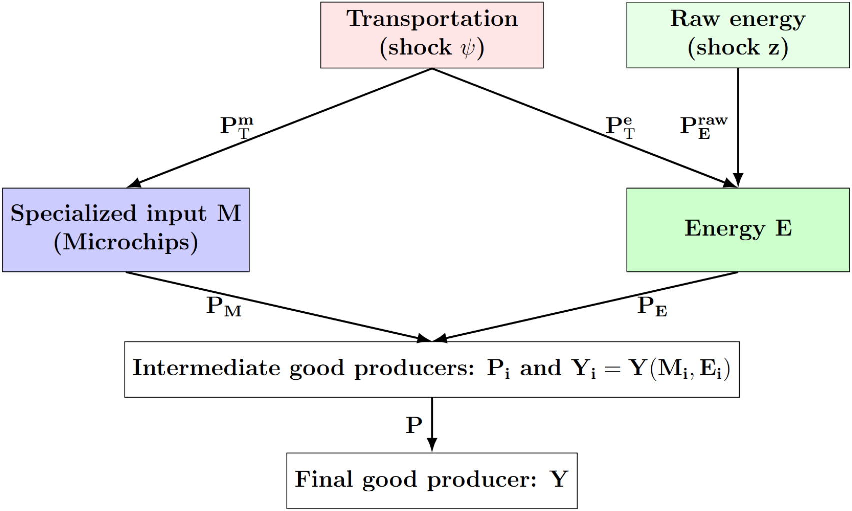

Transportation shock ψ_t hits both inputs: Energy and Microchips (EV batteries)

Production of Final and Intermediate Goods

- **Final good:** CES aggregate of differentiated varieties ($\epsilon > 1$)

$$Y_t = \left(\int_0^1 Y_{i,t}^{\frac{\epsilon-1}{\epsilon}} \, di \right)^{\frac{\epsilon}{\epsilon-1}}$$

- **Intermediate firms:** monopolistically competitive, two inputs

$$Y_{i,t} = M_{i,t}^{\alpha_M} \, E_{i,t}^{\alpha_E}, \qquad \alpha_M + \alpha_E = 1$$

  - **Specialized input** M: non-substitutable (e.g., EV batteries)
  - **Energy** E: commoditized, traded in liquid spot markets

- **Nominal marginal cost** (common across firms):

$$MC_t = \left(\frac{P_{M,t}}{\alpha_M} \right)^{\alpha_M} \cdot \left(\frac{P_{E,t}}{\alpha_E} \right)^{\alpha_E}$$

Energy Production

- Energy prod. function with **IRS** ($\nu > 1$) and high **complementarity** ($\eta \to 0$) of raw energy and transportation:

$$E_t = \left((1-\delta)^{1/\eta} \underbrace{E_{T,t}^{1-1/\eta}}_{\text{energy transportation}} + \delta^{1/\eta} \underbrace{E_{\text{raw},t}^{1-1/\eta}}_{\text{raw energy}} \right)^{\frac{\nu}{1-1/\eta}}$$

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- **Optimal demand for energy transportation**:

$$E_{T,t} = \nu^{\eta} E_t^{\eta + \frac{1}{\nu}(1-\eta)} \cdot (1-\delta) \left(\frac{P^e_{T,t}}{P_{E,t}} \right)^{-\eta}$$

- **Optimal demand for raw energy**:

$$E_{raw,t} = \nu^{\eta} E_t^{\eta + \frac{1}{\nu}(1-\eta)} \cdot \delta \cdot \left(\frac{P^{raw}_{E,t}}{P_{E,t}} \right)^{-\eta}$$

Equilibrium Energy Price

- **Equilibrium supply of energy**:

$$E_t = (\nu P_{E,t})^{\frac{\nu}{1-\nu}} \left[(1-\delta)(P^e_{T,t})^{1-\eta} + \delta(P^{raw}_{E,t})^{1-\eta}\right]^{\frac{\nu}{(1-\eta)(\nu-1)}}$$

Equilibrium price of energy (in logs) :

$$p_{E,t} = \underbrace{(1-\nu)\, y_t}_{\text{IRS effect}} + \nu \Big[\underbrace{(1-\delta)\,\psi_t}_{\text{transportation shock}} + \underbrace{\delta\, z_t}_{\text{raw energy shock}} \Big]$$

  - ψ_t: **true transportation shock** (AR(1), persistence ρ_ψ)
  - z_t: i.i.d. raw energy shock

Equilibrium Specialized Input Price

- **Specialized input** has production structure **symmetrical** to the energy input (raw input + transportation)

- Log-linear equilibrium price of specialized input:

$$p_{M,t} = \underbrace{(1-\nu)\, y_t}_{\text{increasing returns}} + \underbrace{\nu\,(1-\delta)\,\psi_t^*}_{\text{latent component}}$$

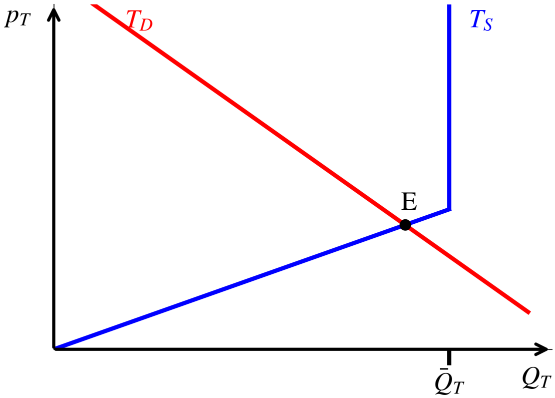
  - ψ_t^*: **estimated transportation shock**

Transportation Market

Four Facts about Global Transportation Network

1. **Maritime dominance.** Over 80% of global merchandise trade moves by sea.
2. **Finite capacity.** Supply is a fixed fleet ($\sim$6,000 container ships) $\Rightarrow$ *vertical kink*: once fully deployed, quantity cannot rise even if demand increases.
3. **Shared chokepoints.** Energy and specialized inputs rely on the same shipping infrastructure $\Rightarrow$ transportation disruptions raise effective delivery costs for all input goods.
4. **Asymmetric exposure.** Energy remains available at a spot premium, while specialized inputs face delivery lags and true supply uncertainty.

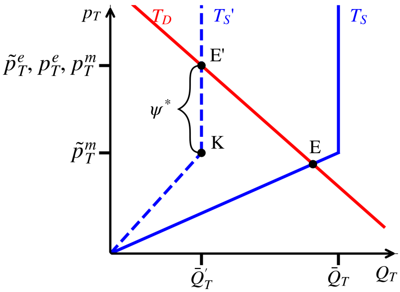
Transportation Market



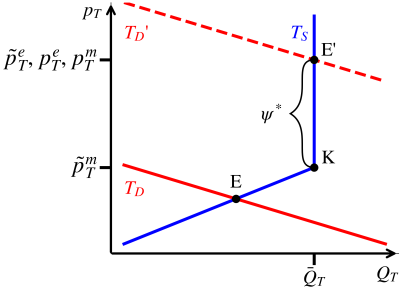
← *capacity constraint*

Unconstrained equilibrium in the transportation market

Shocks in Transportation Market

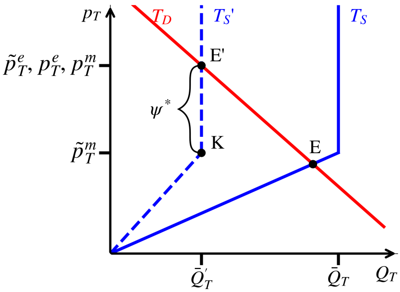


"Strait of Hormuz" congestion supply shock



"Covid" demand shock

Shocks in Transportation Market



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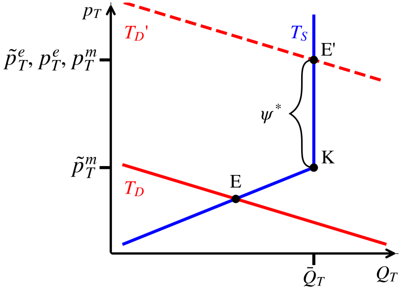

"Covid" demand shock

$$\underbrace{p^m_{T,t}}_{\text{effective price}} = \underbrace{\tilde{p}^m_{T,t}}_{\text{market price}} + \underbrace{\psi^*_t}_{\text{unobserved delay}}$$

Wedge ψ^*_t btw *effective* and *market* **transportation price** of input M

Transportation Prices

- Energy

$$\underbrace{p^{e}_{T,t}}_{\text{effective price}} = \underbrace{\tilde{p}^{e}_{T,t}}_{\text{market price}} = \underbrace{\psi_t}_{\text{true transportation shock}}$$

- Specialized input : unobserved delay ψ^*_t creates **wedge** btw. *effective* and *market* price of transportation

$$\underbrace{p^{m}_{T,t}}_{\text{effective price}} = \underbrace{\tilde{p}^{m}_{T,t}}_{\text{market price}} + \underbrace{\psi^*_t}_{\textbf{unobserved delay}}$$

State-Space Representation of Information Problem

$$
\begin{cases}
\underbrace{\psi_t}_{\substack{\textbf{state: transportation}\\ \textbf{shock}}} = \underbrace{\rho_\psi \, \psi_{t-1}}_{\text{AR(1) persistence}} + \underbrace{\varepsilon_{\psi,t}}_{\substack{\textbf{state noise}\\ \varepsilon_{\psi,t}\sim\mathcal{N}(0,\sigma_\psi^2)}} \\[2ex]
\underbrace{p_{E,t}}_{\substack{\textbf{observation:}\\ \textbf{energy price}}} = \underbrace{(1-\nu)\, y_t}_{\text{output}} + \underbrace{\nu\, \Omega_t}_{\textbf{signal}} \\[2ex]
\underbrace{\Omega_t}_{\textbf{signal}} \equiv \underbrace{(1-\delta)\, \psi_t}_{\substack{\textbf{transportation}\\ \textbf{shock}}} + \underbrace{\delta\, z_t}_{\substack{\textbf{noise (raw energy)}\\ z_t\sim\mathcal{N}(0,\sigma_Z^2)}}
\end{cases}
$$

- **State**: ψ_t captures transportation delays / congestion (AR(1) with variance σ_ψ^2).
- **Observation**: $p_{E,t}$ is a noisy function of the state via Ω_t, plus an output term $(1-\nu)y_t$.
- **Signal vs noise**: Ω_t mixes the true transport shock with raw energy noise.

Kalman Filter: Recursive Learning

- Firms form a **Bayesian estimate** of the transportation delay ψ_t^* from **observed** energy prices.
- Price of energy p_E **noisy signal** of **true** transportation shock
- Updating is **recursive**

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Update (in logs)

$$\underbrace{\psi_t^*}_{\text{posterior}} = \underbrace{\mathbb{E}_{\mathcal{P}_{\psi,(t-1)}}\{\psi_t\}}_{\text{prior}} + \underbrace{\mathbb{K}(\mathcal{S})}_{\text{Kalman gain}} \cdot \underbrace{\left(p_{E,t} - \mathbb{E}_{\mathcal{P}_{\psi,(t-1)}}\{p_{E,t}\}\right)}_{\text{error correction with new signal}}$$

Kalman Filter: Recursive Learning

- Firms form a **Bayesian estimate** of the transportation delay ψ_t^* from **observed** energy prices.
- Price of energy p_E **noisy signal** of **true** transportation shock
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$$\mathcal{S} \equiv \frac{\sigma_\psi^2}{\sigma_z^2} \quad \text{(signal-to-noise ratio)}$$

$\mathcal{S}$ increasing in the variance of the transportation shock (σ_ψ^2).

Real Marginal Cost Under Uncertainty

- Price of **specialized input** M

$$p_M(\psi_t^*) = (1-\nu)\, y_t + \nu(1-\delta) \cdot \underbrace{\Big[\mathbb{E}_{\mathcal{P}_{\psi,(t-1)}}\{\psi_t\} + \mathbb{K}(\mathcal{S})\big(p_{E,t} - \mathbb{E}_{\mathcal{P}_{\psi,(t-1)}}\{p_{E,t}\}\big)\Big]}_{\substack{\psi_t^* \\ \equiv \text{ estimated delay}}}$$

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- Real **marginal cost**

$$mc_t = \alpha_E \cdot \underbrace{p_{E,t}}_{\text{energy price}} + \alpha_M \underbrace{[(1-\nu)\, y_t + \nu(1-\delta)\,\psi_t^*]}_{\text{specialized input price: } p_M(\psi_t^*)}$$

Impact Pass-Through Decomposition

Pass-through of energy prices on final prices

$$\frac{dp_t}{dp_{E,t}} = \underbrace{\alpha_E}_{\text{direct}} + \underbrace{\alpha_M\,\nu\,(1-\delta)\,\mathbb{K}(\mathcal{S})}_{\text{uncertainty effect}}$$

Direct effect $=$ energy cost share α_E Uncertainty effect $\propto \mathbb{K}(\mathcal{S})$

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Direct effect $=$ energy cost share α_E

Uncertainty effect $\propto \mathbb{K}(\mathcal{S})$

Intuition

- Under **incomplete information** ($\mathbb{K}(\mathcal{S}) \neq 0$), marginal costs rise **beyond** the direct energy-cost share α_E.
- Effect scales with $\alpha_M =$ share of **input M** in interm. goods production, and $1-\delta =$ share of **input T** in energy production

Dynamic Pass-Through

Dynamic Pass-through

$$\underbrace{\frac{d\,\mathbb{E}_t\{p_{t+j}\}}{dp_{E,t}}}_{\text{dynamic pass-through}} = \underbrace{\alpha_M\,\nu\,(1-\delta)\,\mathbb{K}}_{\text{impact under uncertainty}} \cdot \underbrace{\left[\rho_\psi\,(1-\mathbb{K}\,\nu\,(1-\delta))\right]^j}_{\text{learning-driven persistence}}$$

- $\mathbb{K} \to 0$: **no persistence** in price response

Dynamic Pass-Through

Dynamic Pass-through

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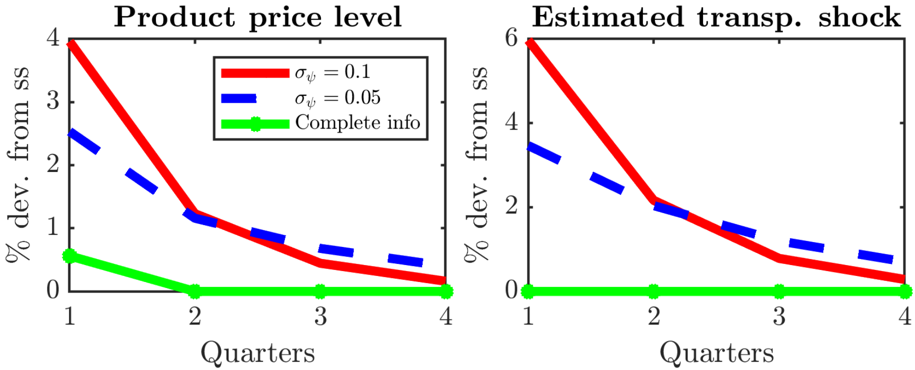
Intuition

Under uncertainty, **persistence** in the transportation shock "contaminates" (*i.i.d.*) energy price shock.

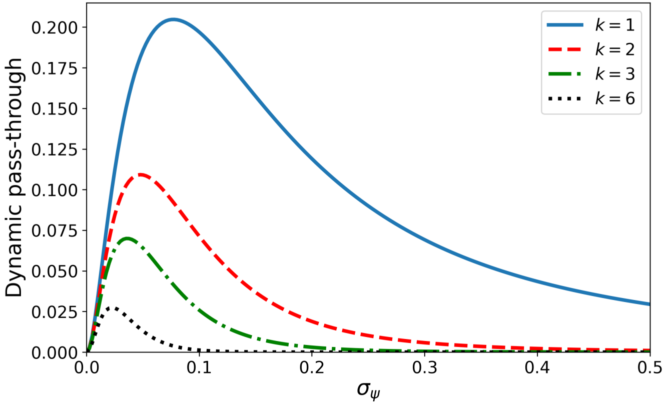
- $\mathbb{K} \to 0$: **no persistence** in price response

Impact Pass-through of Energy to Prices



Effect of *iid* **energy price** shock

Effect of Uncertainty on Dynamic Pass-Through



Bell-shaped effect of transportation uncertainty σ_ψ on dynamic pass-through at horizon k

Nominal Rigidities and NKPC

General Equilibrium NK Model with Supply Chain Uncertainty

- Extend to **general equilibrium**: households + firms + supply chain
- **Labor** in production function + labor supply
- **Calvo pricing** for intermediate goods firms → Derive **NKPC under supply chain uncertainty**

⇒ **Key result:** uncertainty generates **endogenous cost-push** component in the NKPC

Endogenous cost-push effect in the Phillips curve

NK Phillips curve with uncertainty

$$\pi_t \;=\; \beta\,\mathbb{E}_t\{\pi_{t+1}\} \;+\; \frac{(1-\theta)(1-\theta\beta)}{\theta}\,mc_t \;+\; \underbrace{G\,u_t}_{\textbf{uncertainty cost-push}}$$

$$u_t \;=\; L\,u_{t-1} \;+\; \mathbb{K}(\mathcal{S})\,\nu\;\underbrace{FE(p_{E,t})}_{\textbf{forecast error}}\,, \qquad L \equiv \left[1-\mathbb{K}(\mathcal{S})\,\nu\,(1-\delta)\right]\rho_\psi$$

Endogenous cost-push effect in the Phillips curve

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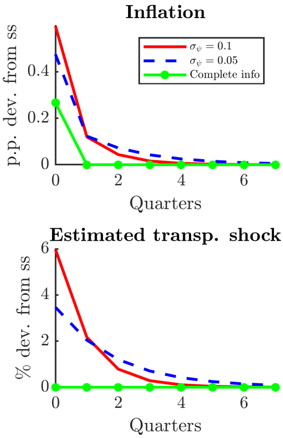
- **Comparative statics:** larger $\mathbb{K} \Rightarrow$ stronger amplification; higher $\mathbb{K} \Rightarrow$ lower L (faster learning, less persistence).
- **Benchmark (complete information)** if $\mathbb{K} = 0$ then $u_t = 0 \rightarrow$ NKPC reverts to **standard** form.

Calibration

	Description	Value
α_N	Share of labor in output production	0.66
α_E	Share of energy in output production	0.03
α_M	Share of specialized input in output production	0.31
ρ_ψ	Persistence of transportation shock	0.90
σ_z	Standard deviation of energy price shock	0.10
δ	Share of raw energy in energy production	0.50
ν	Increasing returns to scale index	1.25
η	Elasticity of substitution between raw energy and transportation	0.01

Table: Key model parameters

IRFs to a raw energy price shock



- **Incomplete information amplifies** and **propagates** the effect of energy price shock on **inflation**

When does look-through fail?

- Under **complete information**, energy shocks raise inflation only temporarily, leaving expectations unchanged.

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- Our mechanism: with **supply-chain uncertainty**, firms imperfectly observe upstream conditions.

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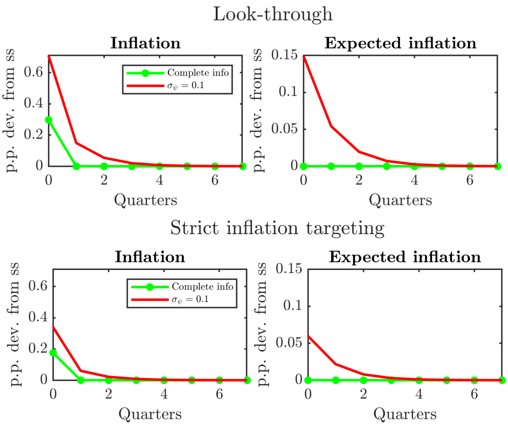
When does look-through fail?

- Under **complete information**, energy shocks raise inflation only temporarily, leaving expectations unchanged.
- Our mechanism: with **supply-chain uncertainty**, firms imperfectly observe upstream conditions.
- A temporary energy shock may then be interpreted as a **signal of persistent future costs**.
- Result: **current inflation rises more** and **expected inflation becomes persistent**.

Takeaway: look-through is harmless only in low-uncertainty environments.

Energy shocks under uncertainty: look-through or respond?

- **Top panel: Look-through** ($\phi_\pi = 1.0001$): inflation and expected inflation become more persistent.
- **Bottom panel: Strict inflation targeting** ($\phi_\pi = 4$): both responses are dampened substantially.



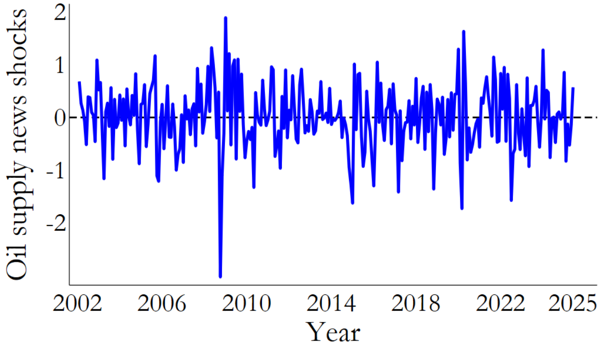
Takeaways

- **SCU** amplifies and propagates effect of energy shocks on **inflation**
- Policy implication (I): Central banks must **condition** responses on SCU, not just shock size.
- Policy implication (II): **Look-through** energy shocks **breaks down** under high SCU

Details on the model

Robustness

Oil supply news shocks



High-frequency identification of oil supply news shocks

Monthly series, 2002–2025

Based on Känzig (2021)

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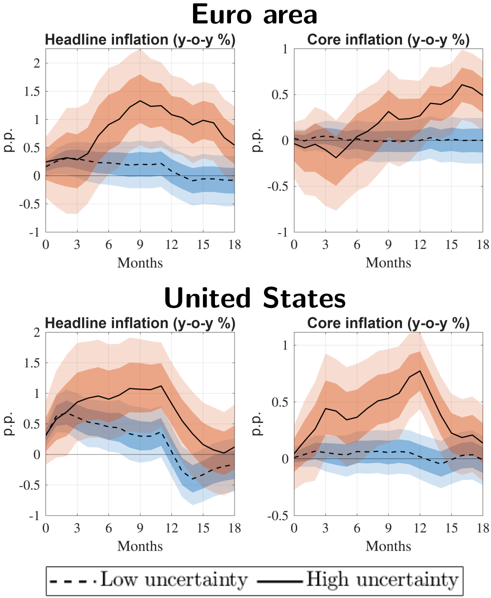
Local projection: Global SCU (top vs bottom 20%)



Using Global SCU

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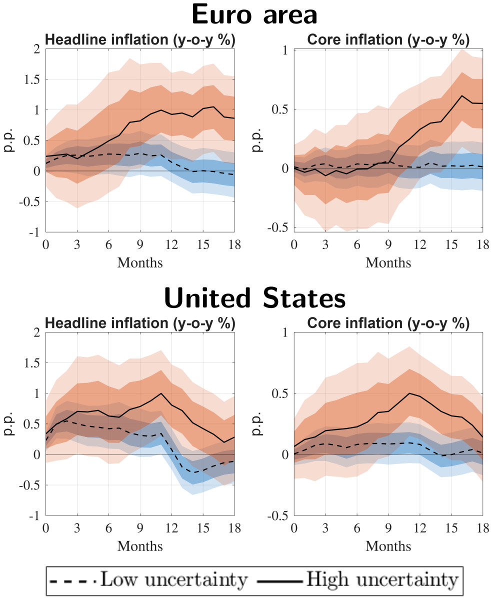
Local projection: ETU index (top vs bottom 20%)



Using ETU index.

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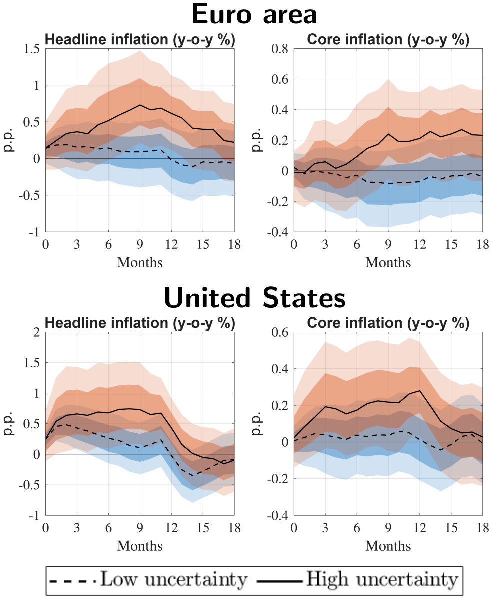
Local projection: Var GSCPI (top vs bottom 20%)



Using (stochastic volatility) Var(GSCPI).

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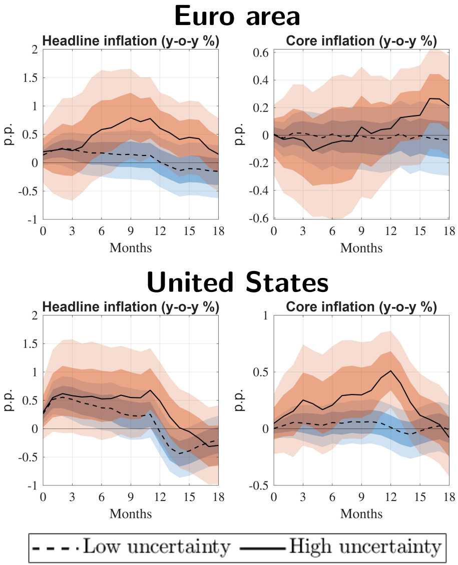
Local projection: Regional SCU (top vs bottom 20%), with Covid-19 inflation dummy



The pass-through of energy to CPI inflation is dependent on supply chain uncertainty, also adding a control for the Covid-19 inflation period.

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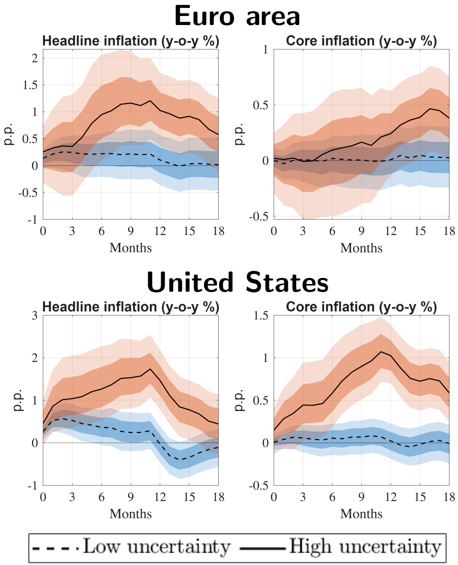
Local projection: Regional SCU (top vs bottom 20%), with Covid-19 closure period inflation dummies



The pass-through of energy to CPI inflation is dependent on supply chain uncertainty, also out of the Covid-19 closure period inflation sample (Apr. 2020 - Dec. 2020).

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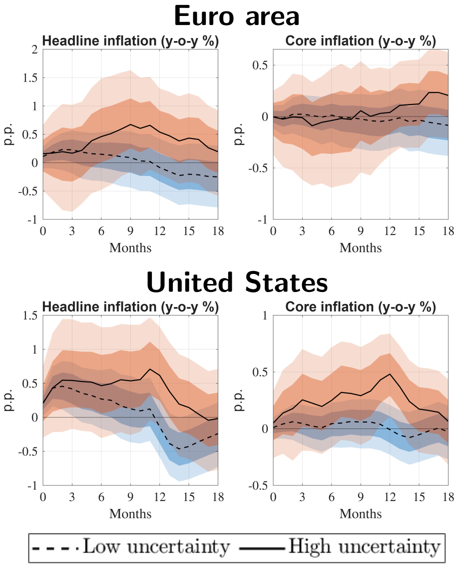
Local projection: Regional SCU (top vs bottom 20%), with additional supply chain pressures controls



Results are broadly unchanged when adding also more controls to the regression equation, namely GSCPI levels and its lags.

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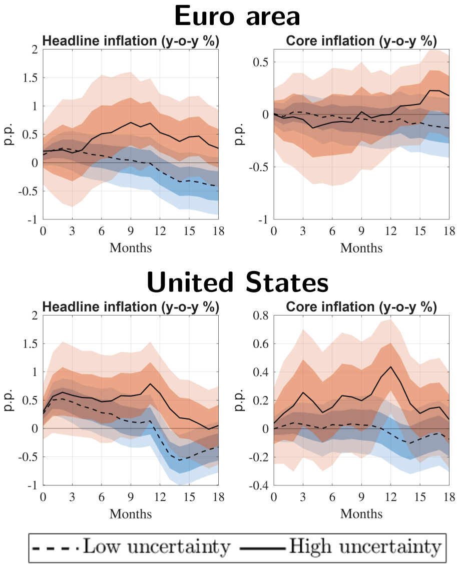
Local projection: Regional SCU (top vs bottom 20%), controlling for the interaction of oil shocks with GSCPI



Same results when controlling for supply-chain pressure (GSCPI) interacted with the oil news shock.

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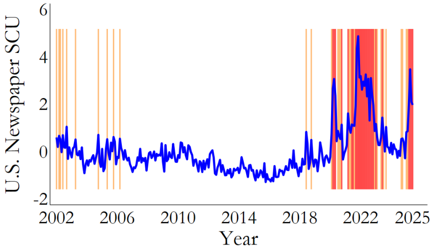
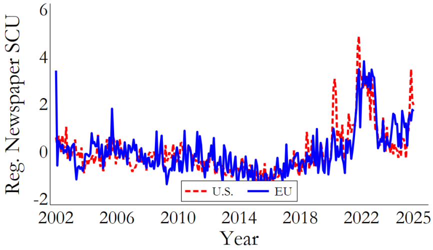
Appendix

US Region-specific SCU



(a) U.S. Region-specific SCU



(b) U.S. vs EU Region-specific SCU

SCU

GSCPI Volatility Index

- Start from the monthly **GSCPI**, a measure of global supply-chain pressures based on transportation costs and PMI survey data.
- Model the standardized GSCPI as an autoregressive process with stochastic volatility:

$$\psi_t = \rho_\psi \psi_{t-1} + \exp(\sigma_{\psi,t})\varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0,1)$$

$$\sigma_{\psi,t} = (1-\rho_\sigma)\bar{\sigma}_\psi + \rho_\sigma \sigma_{\psi,t-1} + \eta_\psi u_t, \qquad u_t \sim \mathcal{N}(0,1)$$

- Estimate the model with **Bayesian methods / particle filtering** and recover the filtered latent volatility state over time.
- Define the **GSCPI Volatility Index** as the filtered conditional variance of innovations to the GSCPI process.

GSCPI Volatility Index

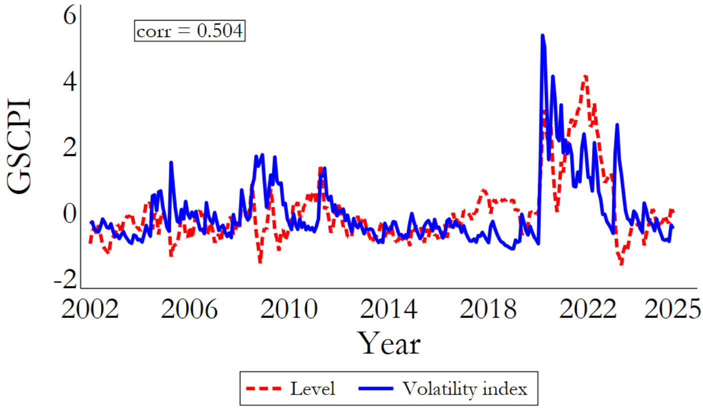
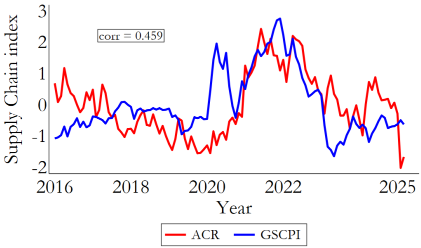


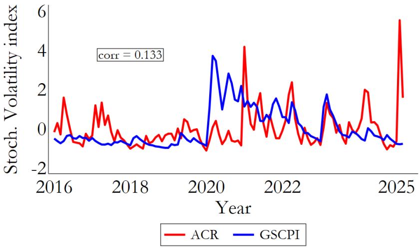
Figure: Evolution of GSCPI: Level versus Volatility

SCU

Average Congestion Rate (ACR) Index



(a) Evolution of GSCPI versus ACR index

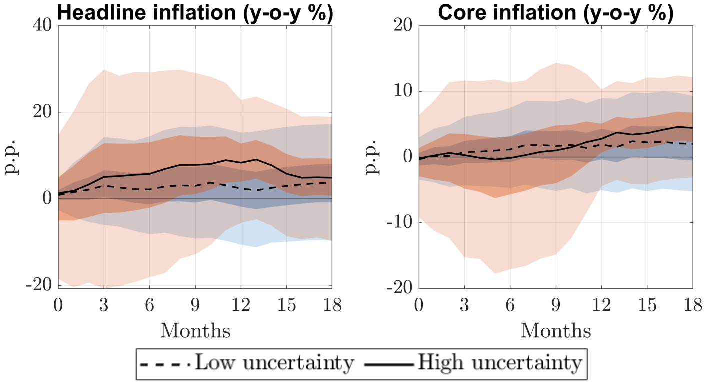


(b) GSCPI vs. ACR volatility index

ACR: measures real-time container ship congestion at major ports worldwide using high-frequency satellite data, capturing physical delivery bottlenecks and shipping disruptions.

SCU

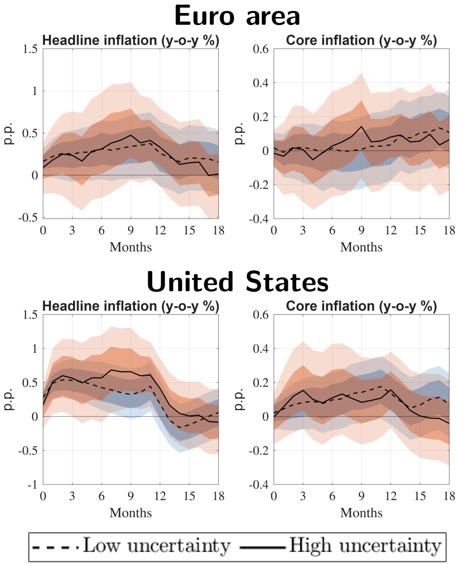
Local projection: Gas Shocks



Pass-through of gas shocks invariant to SCU ⇒ Gas travels via pipelines in regional markets rather than via maritime routes.

LP

Local projection: Economic Policy Uncertainty (EPU)



Amplification of energy shocks is not a general feature of uncertainty.

LP

Kalman filter procedure estimates

Parameter	Low uncertainty	High uncertainty
Energy loading α_1	0.20	1.53
	(5.64×10^{-2})	(0.42)
BDI loading α_2	0.99	8.66×10^{-2}
	(0.19)	(7.76×10^{-2})
Persistence ρ_ψ	0.95	0.92
	(1.99×10^{-2})	(4.49×10^{-2})
Obs. var. (energy) v_1	0.43	1.15×10^{-9}
	(4.07×10^{-2})	(8.51×10^{-8})
Obs. var. (BDI) v_2	5.03×10^{-10}	0.40
	(2.01×10^{-8})	(7.54×10^{-2})
State var. σ_ψ^2	0.11	0.17
	(4.23×10^{-2})	(9.26×10^{-2})

Table: Kalman parameters across uncertainty regimes in the Euro area

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Kalman filter procedure estimates

Parameter	Low uncertainty	High uncertainty
Energy loading α_1	0.34 (8.75×10^{-2})	1.21 (0.34)
BDI loading α_2	0.99 (0.21)	0.27 (9.58×10^{-2})
Persistence ρ_ψ	0.96 (1.80×10^{-2})	0.91 (5.04×10^{-2})
Obs. var. (energy) v_1	0.68 (6.40×10^{-2})	1.88×10^{-8} (1.11×10^{-6})
Obs. var. (BDI) v_2	8.05×10^{-10} (3.78×10^{-8})	0.23 (4.29×10^{-2})
State var. σ^2_ψ	9.39×10^{-2} (3.86×10^{-2})	0.19 (0.10)

Table: Kalman parameters across uncertainty regimes in the U.S.

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Kalman gains for inflation

Signal	Low uncertainty	High uncertainty
Energy inflation	4.09×10^{-10}	0.83
BDI	1.01	1.51×10^{-8}

(a) U.S.

Signal	Low uncertainty	High uncertainty
Energy inflation	2.40×10^{-10}	0.65
BDI	1.01	1.06×10^{-10}

(b) Euro area

Table: Kalman gain for BDI and Energy inflation - U.S. and Euro area

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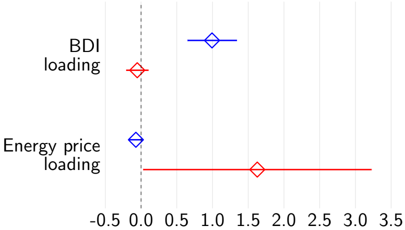
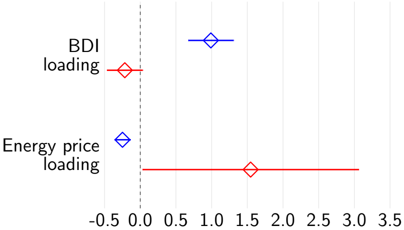
Kalman filter: loading coefficients for prices



(a) U.S.



(b) Euro area

High supply-chain uncertainty   Low supply-chain uncertainty

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Kalman gains for prices

Signal	Low uncertainty	High uncertainty
Energy price	-2.71 $\times 10^{-10}$	0.61
BDI	1.01	-3.28 $\times 10^{-8}$

(a) U.S.

Signal	Low uncertainty	High uncertainty
Energy price	-4.30 $\times 10^{-10}$	0.65
BDI	1.01	-6.85 $\times 10^{-4}$

(b) Euro area

Table: Kalman gain for BDI and Energy prices - U.S. and Euro area

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SCU from Earnings Calls

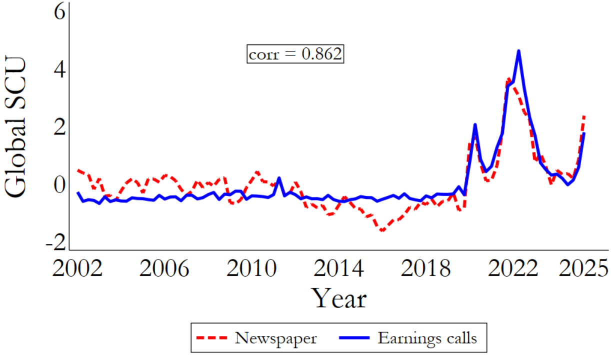

Figure: Global SCU index extracted from newspapers versus earnings calls

SCU